

Digital Circuits

Power Analysis

©Hanan Ribo

Static Power Dissipation

CMOS Static Power Dissipation

In CMOS circuits, static power dissipation P_{static} is the power consumed when the circuit is in a stable, idle state (*non-switching*), and primarily caused by leakage currents.

The static power dissipation:

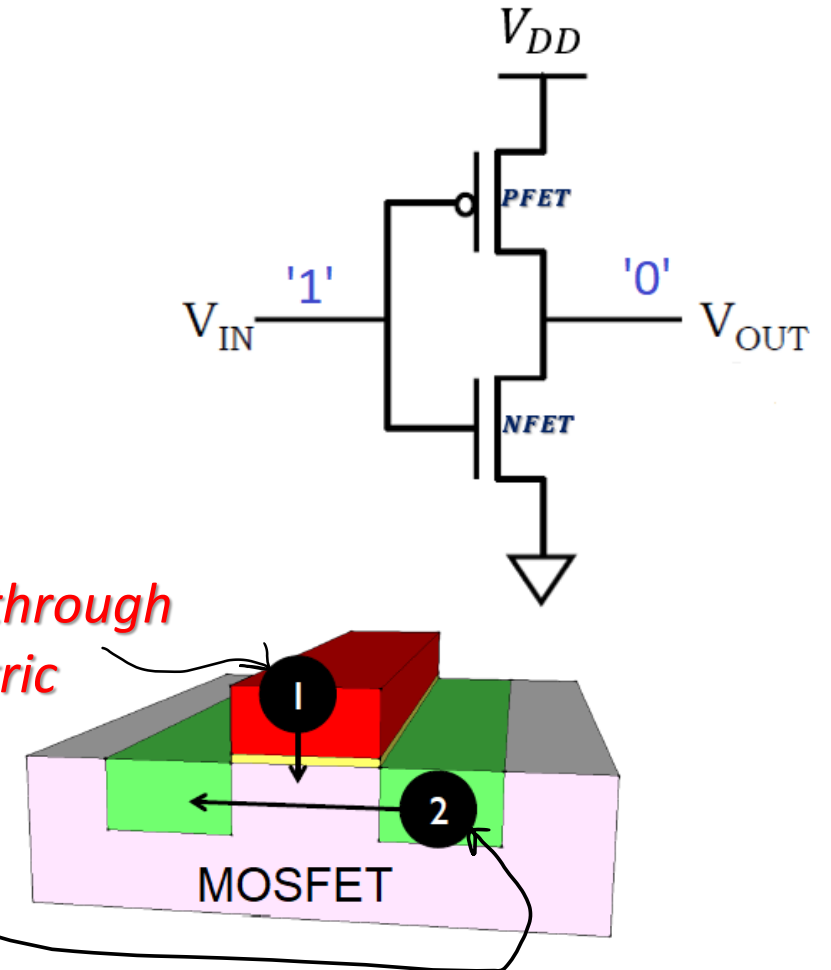
$$P_{static} = I_{leakage} \cdot V_{DD}$$

The static power dissipation through N-transistors:

$$P_{static} = V_{DD} \sum_{i=1}^N I_{leakage_i}$$

Current leakage from drain to source even though MOSFET is “off” (subthreshold conduction)

tunneling electrons through the gate dielectric



Total DC leakage current

The total DC leakage current $I_{leakage} = I_{sub} + I_{ox} + I_{GIDL}$ includes:

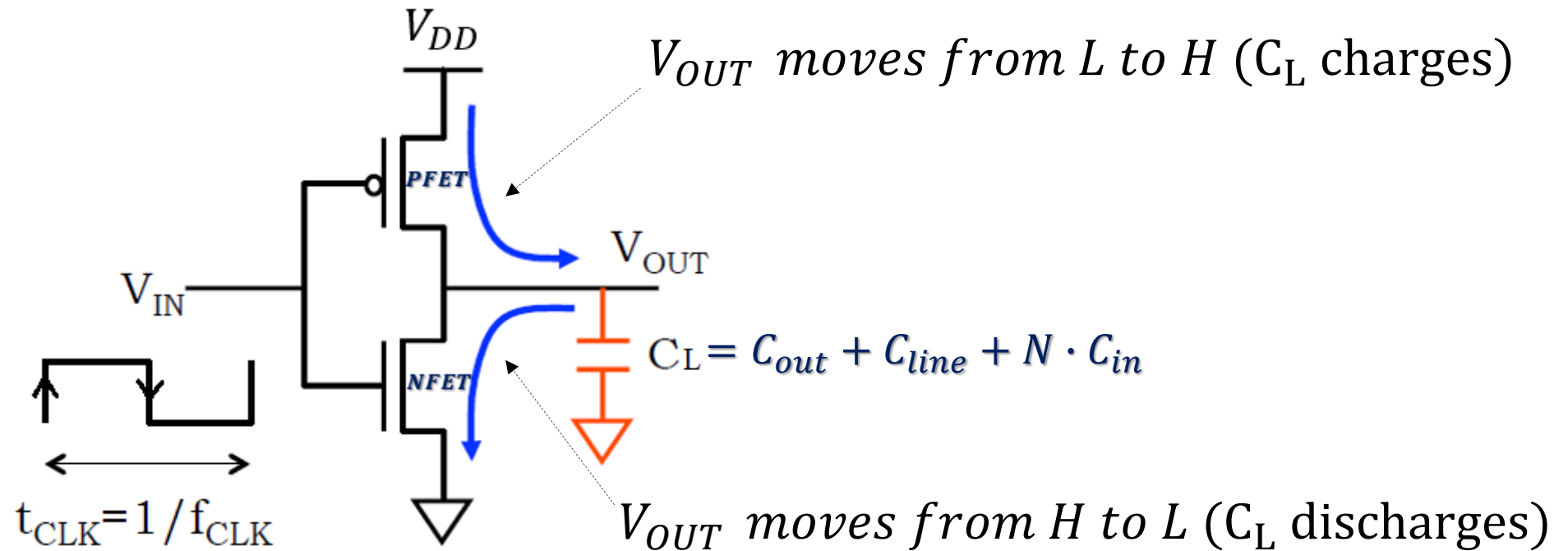
- 1. Subthreshold** leakage I_{sub} flowing from the drain to the source
when $V_{GS} < V_{th}$, depends strongly on temperature and V_{th} (gets larger as difference between V_{th} and V_{Goff} gets smaller)
- 2. Gate Oxide** leakage I_{ox} tunneling through the gate dielectric (dependent on the gate oxide thickness and material)
- 3. Junction** leakage I_{GIDL} caused by high electric fields in the drain junction (dependent on the voltage difference between the gate and drain)

Dynamic Power Dissipation

CMOS Dynamic Power Dissipation

Dynamic Power is consumed when transistors actively switch states.

Energy is drawn from the supply to charge/discharge node capacitances



$$P_{out} = I_{out} \cdot V_{out} = C \cdot \frac{d(V_{out})}{dt} \cdot V_{out}$$

$$E_{discharge} = -C \int_0^{T_{clk}} \frac{d(V_{out})}{dt} \cdot V_{out} dt = -C \int_0^{T_{clk}} V_{out} dV_{out} = C \frac{V_{out}^2}{2}$$

$$E_{charge} = C \int_{\frac{T_{clk}}{2}}^{T_{clk}} \frac{d(V_{out})}{dt} \cdot V_{out} dt = C \int_{\frac{T_{clk}}{2}}^{T_{clk}} V_{out} dV_{out} = C \frac{V_{out}^2}{2}$$

$$E_{dym} = \int_0^{T_{clk}} P_{out}(t) dt = C \cdot V_{DD}^2 \text{ [Wsec]}$$

$$P_{dym} = f_{clk} \cdot E_{dym} = f_{clk} \cdot C \cdot V_{DD}^2 \text{ [W]}$$

Energy vs. Power

Energy:

- The total capacity to do work (is a measure of quantity), like a car's total fuel *or* like battery total capacity.
- Energy measure unit is joules **$[J] = [Wsec]$** .

Power:

- The rate at which that work is done over time (the speed the energy is used), like a car's engine horsepower or battery electrical power.
- Power measure unit is watts **$[W] = [J/sec]$** (i.e., energy per second).

Electrical Power and Energy:

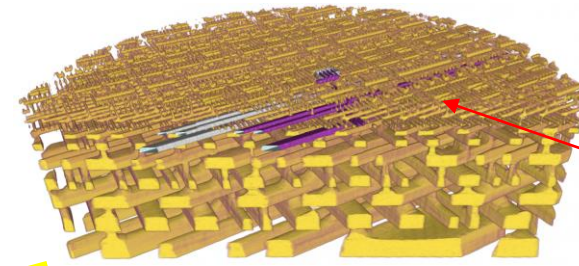
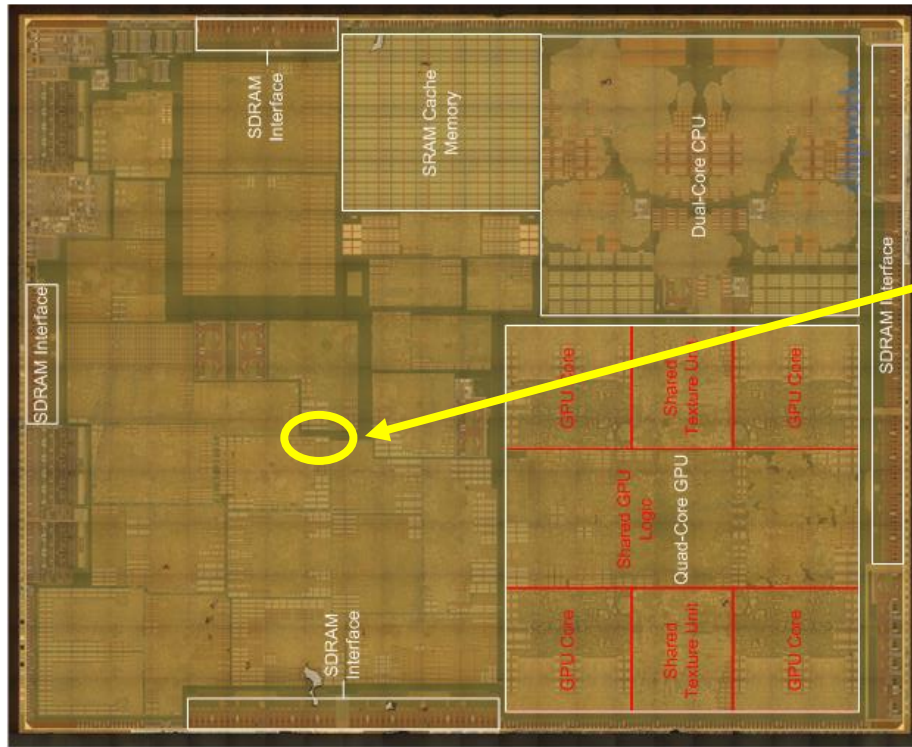
$$P = I \cdot V [W] , \quad E = I \cdot V \cdot t [J]$$

How much energy a 100W light bulb running for 2 hours uses ?

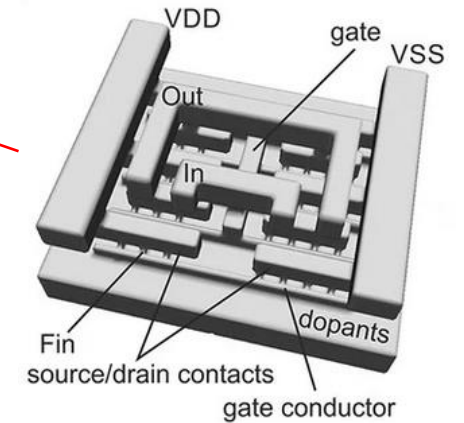
$$E = 200 Wh = 200 \cdot 3600 J = 720,000 J$$



CMOS Dynamic Power Dissipation



per chip



per node

How can we estimate the dynamic power dissipation per chip?

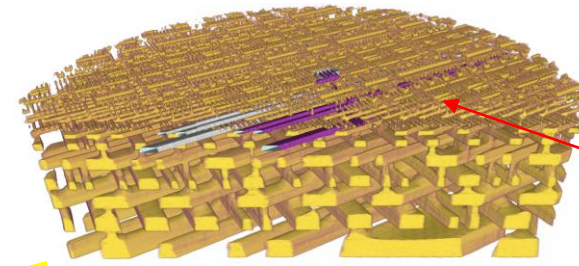
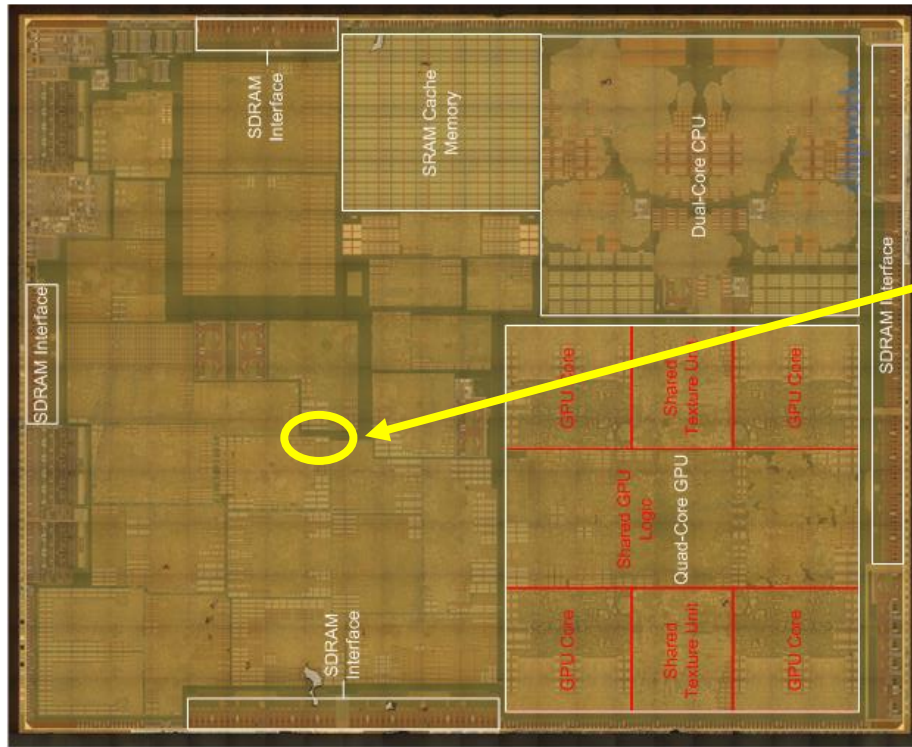
f_{CLK} = transitions (charge **or/and** discharge) frequency

N = number of changing nodes

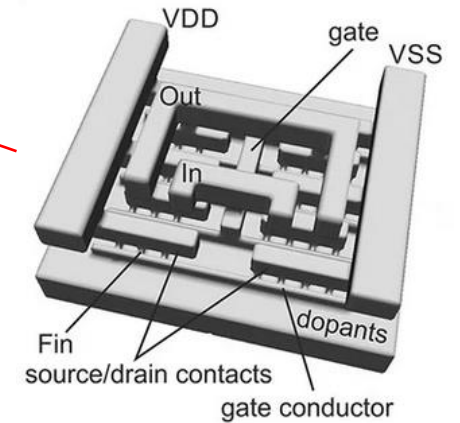
Power dissipation (per node) = $f_{CLK} \cdot C \cdot V_{DD}^2$

Power dissipation (per chip) = $f_{CLK} \cdot N \cdot C \cdot V_{DD}^2$

CMOS Dynamic Power Dissipation - Example



per chip



per node

$$f_{CLK} \cong 1GHz = 10^9 \left[\frac{cycles}{sec} \right], N \cong 10^8 \left[\frac{changing\ nodes}{cycles} \right], C_{out} \cong 1[fF] = 10^{-15} \left[\frac{Farad}{node} \right], V_{DD} \cong 1V$$

Estimation of the chip dynamic power dissipation?

$$Dynamic\ Power\ dissipation = f_{CLK} \cdot N \cdot C \cdot V_{DD}^2 = 10^9 \cdot 10^8 \cdot 10^{-15} \cdot 1^2 = 100W$$

Counter layout on an FPGA - example

Power Analyzer Summary	
<<Filter>>	
Total Thermal Power Dissipation	245.30 mW
Core Dynamic Thermal Power Dissipation	11.21 mW
Core Static Thermal Power Dissipation	103.40 mW
I/O Thermal Power Dissipation	130.69 mW

